

Stony Brook University
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ASSIGNMENT 2

Due: February 21, 2018, in class

INSTRUCTIONS: Show all your work to get full points. Please also cut and paste at the end of your submission the R code you have used in problem 4 to show your work.

1. Suppose in a survey of $N = 400$ students, 215 responded that they prefer small classes and 185 responded that they prefer large classes. Let p denote the fraction of all students who preferred small classes at the time of the survey, and $\hat{p}$ be the fraction of survey respondents who preferred small classes. (Hint: X is distributed as a Bernoulli random variable)

- Show that $E(\hat{p}) = p$ and $\text{Var}(\hat{p}) = p(1 - p)/N$.
- Use the survey result to estimate p , and calculate the standard error of your estimator. (Hint: Notice that this is the same as estimating the sample mean)
- What is the p-value for the test $H_0 : p = 0.5$ vs. $H_1 : p \neq 0.5$? Will you reject the null at the 5% significance level?
- Construct a 95% confidence interval for p .

2. Let Y_1, Y_2, Y_3 and Y_4 be independent, identically distributed random variables from a population with mean μ and variance σ^2 . Let $\bar{Y} = \frac{1}{4}(Y_1 + Y_2 + Y_3 + Y_4)$ denote the average of these four random variables.

- What are the expected value and variance of $\bar{Y}$ in terms of μ and σ^2
- Now, consider a different estimator of μ :

$$W = \frac{1}{8}Y_1 + \frac{1}{8}Y_2 + \frac{1}{4}Y_3 + \frac{1}{2}Y_4$$

Show that W is also an unbiased estimator of μ . Find the variance of W .

- Based on your answers to the previous questions, which estimator do you prefer $\bar{Y}$ or W ?

3. You are hired by the governor to study whether a tax on liquor has decreased average liquor consumption in your state. You are able to obtain, for a sample of individuals selected at random, the difference in liquor consumption (in ounces) for the years before and after the tax. For person i who is sampled randomly from the population, Y_i denotes the change in liquor consumption. Treat these as a random sample from a $\text{Normal}(\mu, \sigma^2)$ distribution.

- The null hypothesis is that there was no change in average liquor consumption. State this formally in terms of μ .
- The alternative is that there was a decline in liquor consumption; state this formally in terms of μ .
- Now, suppose your sample size is $n = 900$, and you obtain estimates $\bar{y} = -32.8$ and $s = 466.4$. Calculate the t statistic for testing H_0 against H_1 ; obtain the p -value for the test. (Because of the large sample size, just use the standard normal distribution.) Do you reject H_0 at the 5% level? At the 1% level?

- (d) Would you say that the estimated fall in consumption is large in magnitude? comment on the practical versus statistical significance of this estimate.
- (e) What has been implicitly assumed in your analysis about other determinants of liquor consumption over the two-year period in order to infer causality from the tax change to liquor consumption?

4. On Blackboard under the Problem Set 2 folder, you can find a zip file **CPS08.zip** which contains a database **CPS08.csv** for full-time, full-year workers age 25–34, with a high school diploma or B.A./B.S. as their highest degree. A detailed description is given in the pdf file **CPS08_Description**. In this exercise, you will investigate the relationship between a worker's age (*age*) and average hourly earnings (*AHE*).

(Hint: If you are looking for general methods to save plots in R, please look at the following webpage <http://www.stat.berkeley.edu/~s133/saving.html>

- (a) Load the dataset into R and call it **data_wage**. What command did you use? (**Do not use the “import dataset” command in Rstudio.**)
- (b) How many observations are there in your sample? How many variables?
- (c) Report the mean and the standard deviation for *AHE*. Also, create an histogram for *AHE*. Would you say that this variable is normally distributed?
- (d) Make a scatter plot of *AHE* against *age* for the entire sample. What do you observe? Do average earnings increase with age or not? Explain.
- (e) Repeat the previous question, but this time split the sample into males and females. Would you say that the relationship between age and *AHE* does not depend on gender? Explain.